

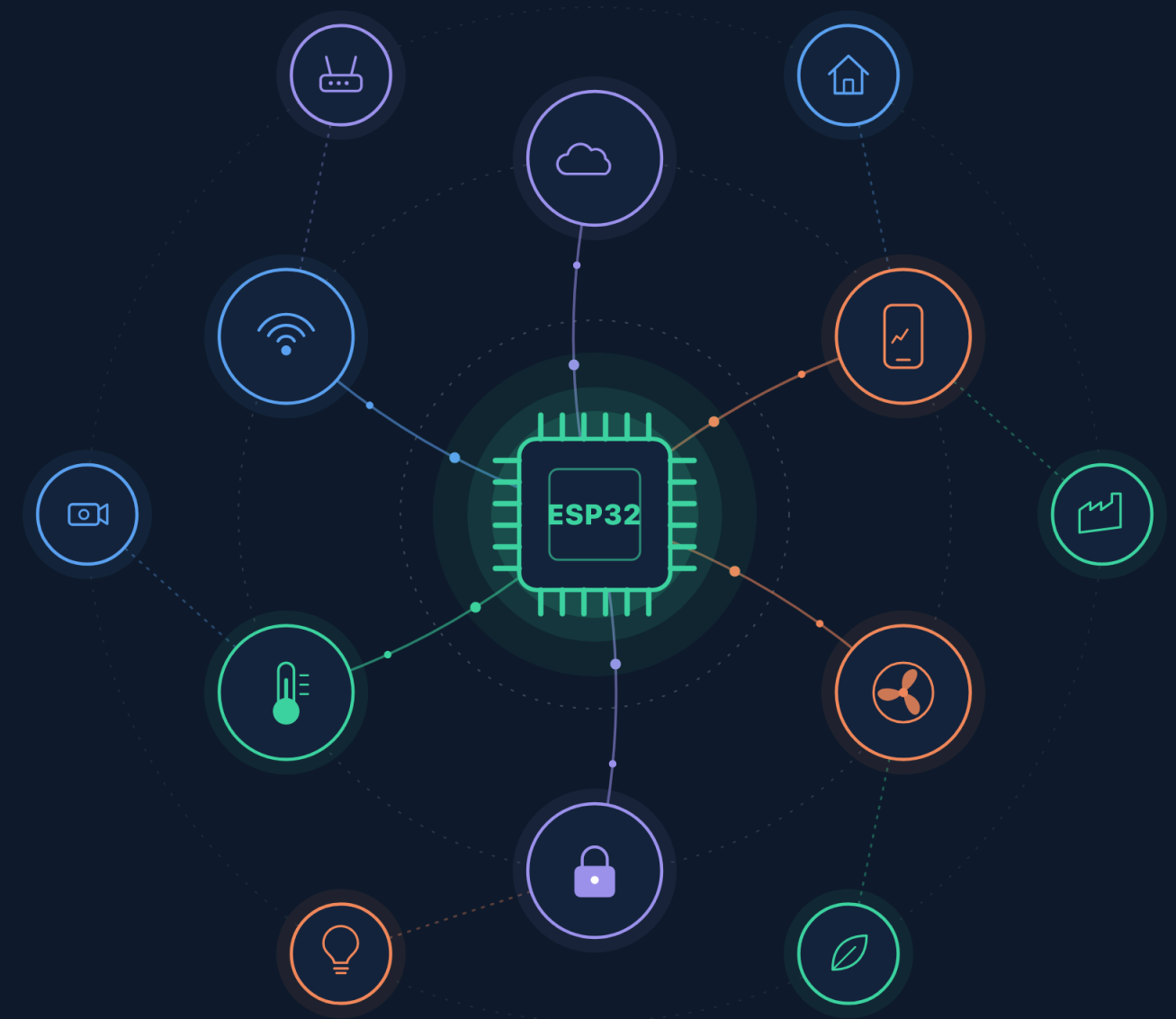
MODULE 5 • 45 MINUTES

IoT Cloud Platforms and Data Visualisation

Where your readings go, how the cloud makes sense of them, and how to show them on a dashboard people understand.

Mehdi Lotfinejad

Introduction to the Internet of Things



THE ROAD AHEAD

Eight modules, one smart device

PART 1 — DEVICES AND DATA

01

What is IoT?

Lecture

02

How it's built

Lecture

03

Sensors and
actuators

Labs 1–2

04

Getting online

Lab 3

PART 2 — CLOUD, AUTOMATION AND SECURITY

05

Cloud and
dashboards

You are here

06

Automation

Lab 5

07

Security

Group activity

08

Real world +
capstone

Lab 6

BY THE END OF THIS MODULE

You will be able to...

-
- 01 **Explain** what the cloud does for an IoT system
 - 02 **Describe** time-series data and four kinds of analytics
 - 03 **Compare** ThingSpeak, Arduino Cloud, Blynk and AWS IoT Core
 - 04 **Choose** the right chart and design a clear dashboard
 - 05 **Build** a live cloud dashboard in Lab 4
-

Part A

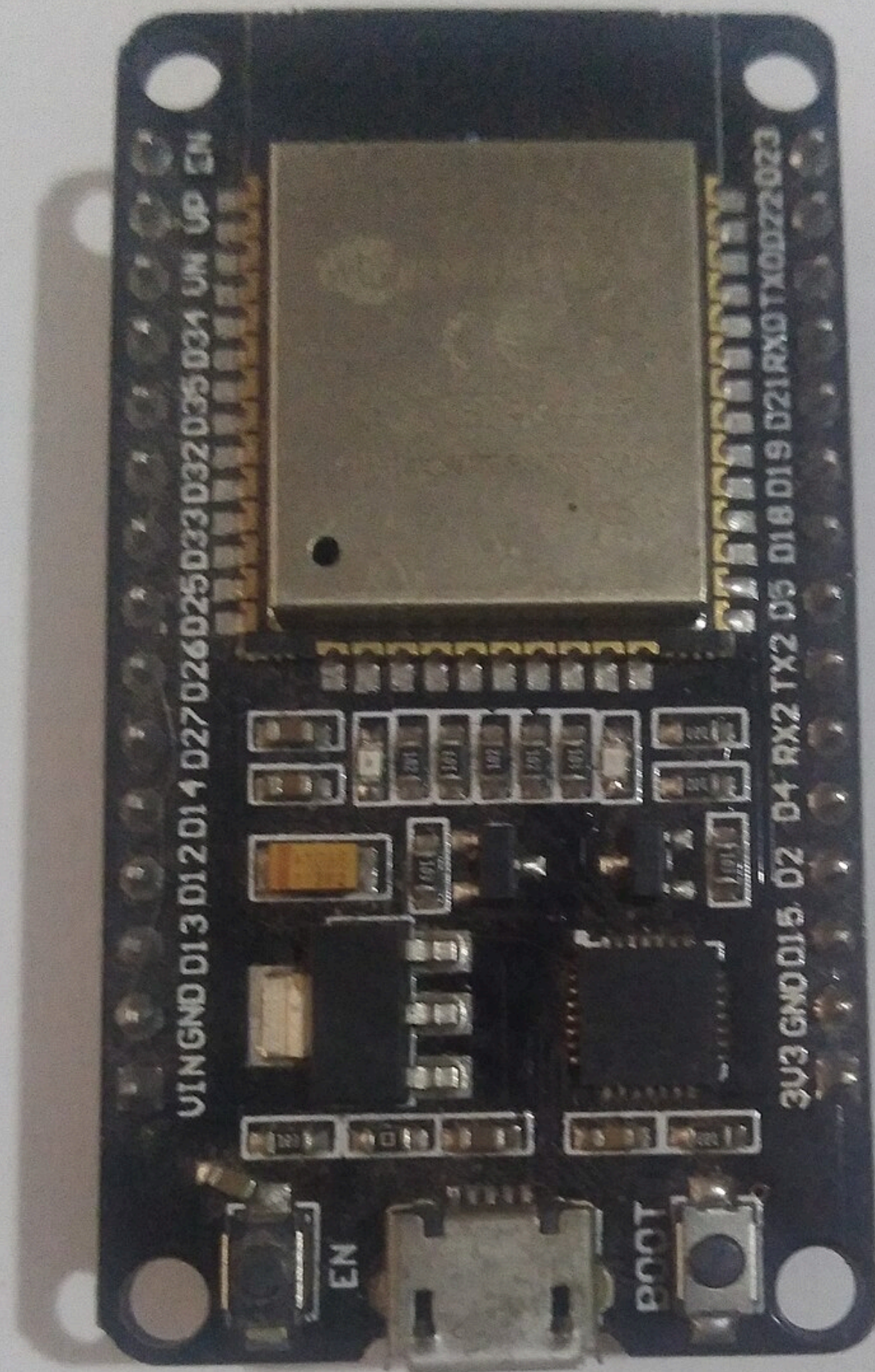
What the cloud does for IoT

In Lab 3, your readings vanished when you closed the tab. Where should they live?

WHAT IS “THE CLOUD”?

Cloud computing means renting computing power, storage and software over the internet.

Like electricity: you don't build a power station to boil a kettle. You plug into the grid and pay for what you use.



WHY IOT NEEDS THE CLOUD

A small board with a tiny memory

An ESP32 is small and cheap on purpose. Could it store years of readings and serve a dashboard to hundreds of people?

What the cloud provides

The cloud provides...	Why it matters for IoT
Storage	Keeps history, so you see trends, not just “right now”
Computing power	Runs analysis far too heavy for a microcontroller
Scale	Handles one device or a million, with no new hardware
Anywhere access	Check your data from any phone or computer
Integration	Connects IoT data to email, billing, maps, AI
Device management	Tracks every device, its status and its permissions

WHAT AN IOT PLATFORM DOES

From device to dashboard



Ingest

Data arrives by MQTT or HTTP.



Manage devices

Credentials, status and settings.



Store

Keep every reading with its time.



Analyse

Averages, trends and alerts.



Visualise

Charts, gauges and dashboards.



Rules and actions

“If this, then that” — in Module 6.

Plus **APIs**: web addresses that let other software read or write your data.

STORING IOT DATA

Time-series data: every value has a time



09:00:00

24.0 °C · 40.0 %



09:00:20

24.1 °C · 40.2 %



09:00:40

24.3 °C · 40.1 %

Without the time, 24.3 °C tells you very little. **With** the time, you can see if the room is getting hotter.

HOW MUCH DATA IS THAT?

One sensor, one reading every 20 seconds

4,300

readings a day

1.6 million

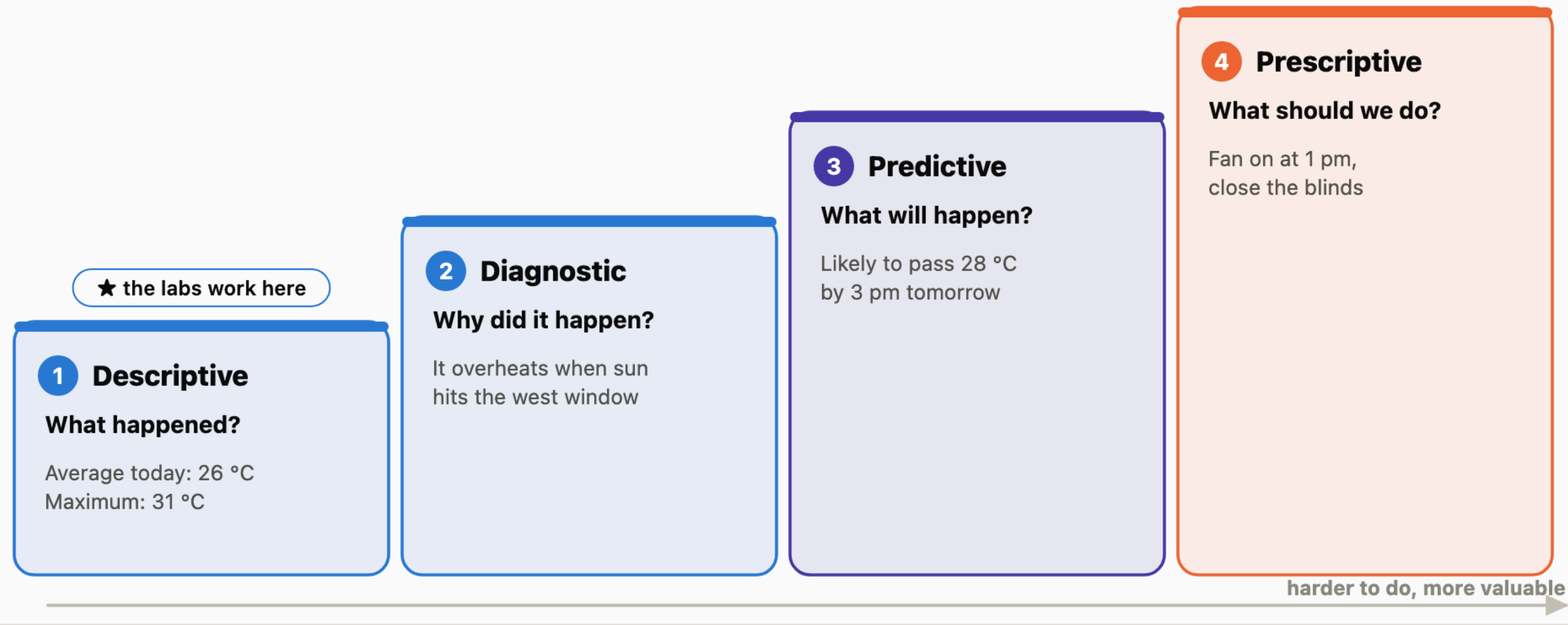
readings a year

Platforms limit **messages** (how many you send) and **retention** (how long data is kept).

Four kinds of analytics

Four levels of analytics

Each step answers a harder — and more valuable — question about the same data.



PAUSE AND CHECK

A farm dashboard shows: “Soil in Field 2 will be too dry in about 6 hours.” Which level of analytics is that?

Descriptive, diagnostic, predictive or prescriptive? Decide with your neighbour.

Part B

Meet the platforms

Four platforms, from the simplest to the most powerful.

ThingSpeak: channels, fields and keys



Channels

One per project or device, like “My Room Monitor”.



Fields

Up to 8 values per channel:
Field 1 = Temperature.



API keys

The **Write** key adds data. The **Read** key reads a private channel.

Free plan: about 3 million messages a year, 4 channels, **15 s** between updates, non-commercial

SIDE BY SIDE

Four IoT platforms compared

Platform	Aimed at	Starting	Strong point	Free option
ThingSpeak	Learners, researchers	Very easy	Simplicity, MATLAB analysis	Yes, non-commercial
Arduino Cloud	Makers, schools	Easy	Code and cloud linked	Yes, very limited
Blynk	App-style prototypes	Easy	Phone control	Yes, limited
AWS IoT Core	Businesses, large scale	Harder	Scale, security, integration	Credits for new accounts

Free-plan details change often. Always check the provider’s pricing page first.

A WORD OF CAUTION

Choosing a platform: four risks



Lock-in

Moving thousands of devices later is painful. MQTT and HTTP help.



Shutdowns

Google Cloud IoT Core shut down in August 2023.



Cost at scale

A free plan for 5 devices says little about 50,000.



Personal data

Data about people must follow laws such as Malaysia's PDPA.

Part C

Dashboards and data visualisation

A good dashboard answers the viewer's questions in seconds.

CHOOSING THE RIGHT CHART

Different questions need different visuals

To show...	Use a...	Example
Change over time	Line chart	Temperature over the last 24 hours
Current value in a range	Gauge or big number	“Tank is 72% full”
A comparison	Bar chart	Energy used by each floor
On/off or good/bad	Status indicator	“Pump: RUNNING”
Where things are	Map	Locations of delivery lorries
Exact values	Table	The last 10 readings, with times

For IoT sensor data, the **line chart** is the workhorse.

Seven rules for a clear dashboard

Rule	In practice
1 Know your audience	A farmer needs “Water Field 3 now”, not raw numbers
2 Most important first	Make it the biggest thing on the screen
3 Always show units	“24” on its own could mean anything
4 Sensible time range	Last hour for “right now”, last week for patterns
5 Show limits	A line for “safe maximum: 28 °C”
6 Not colour alone	Pair red and green with words, like “HIGH” or “OK”
7 Less is more	If nobody acts on a chart, remove it

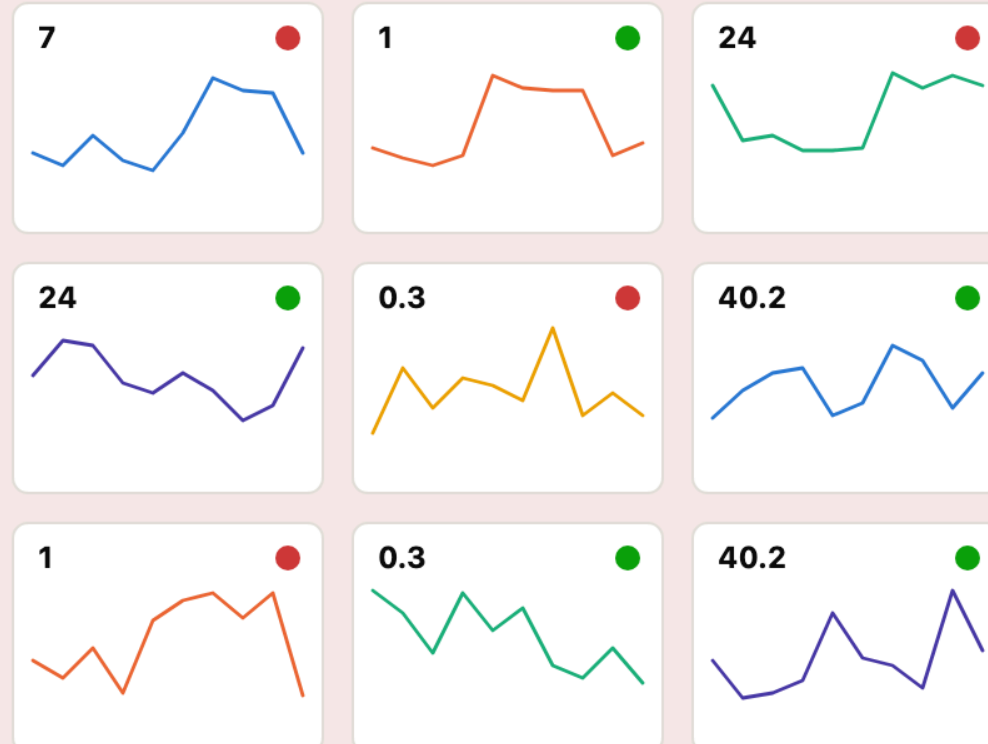
THE SAME DATA, TWO DASHBOARDS

The rules in action

The same data, two dashboards

A good dashboard answers “is everything OK, and what should I do?” in a few seconds.

✗ Hard to read



✗ no units

✗ colour-only status

✗ no clear priority

✗ no limits shown

✓ Easy to read

Living room monitor

2 1 h 24 h 7 d

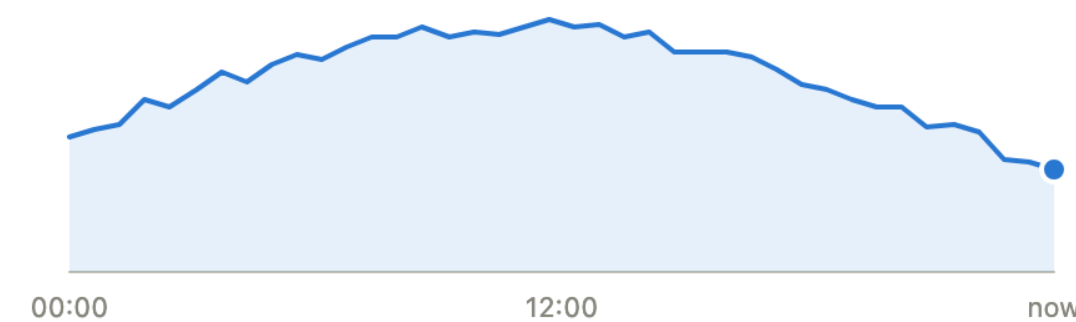
Temperature

Humidity

1 26.4 °C ✓ OK

58 %

3 safe maximum 28 °C



4 ✓ No alerts in the last 24 hours

① number + unit + label ② time range ③ limit shown ④ plain words

PAUSE AND CHECK

A hospital wants one screen showing whether each of 20 medicine fridges is in its safe range. Which visual would you choose, and what would you avoid?

Use the chart table and the seven rules. Discuss in pairs for one minute.

COMMON MISCONCEPTIONS

Four ideas that trip people up

✗ Myth *The cloud is one big computer somewhere.*

✓ Fact It's many computers in data centres around the world.

✗ Myth *In the cloud, data is automatically safe and permanent.*

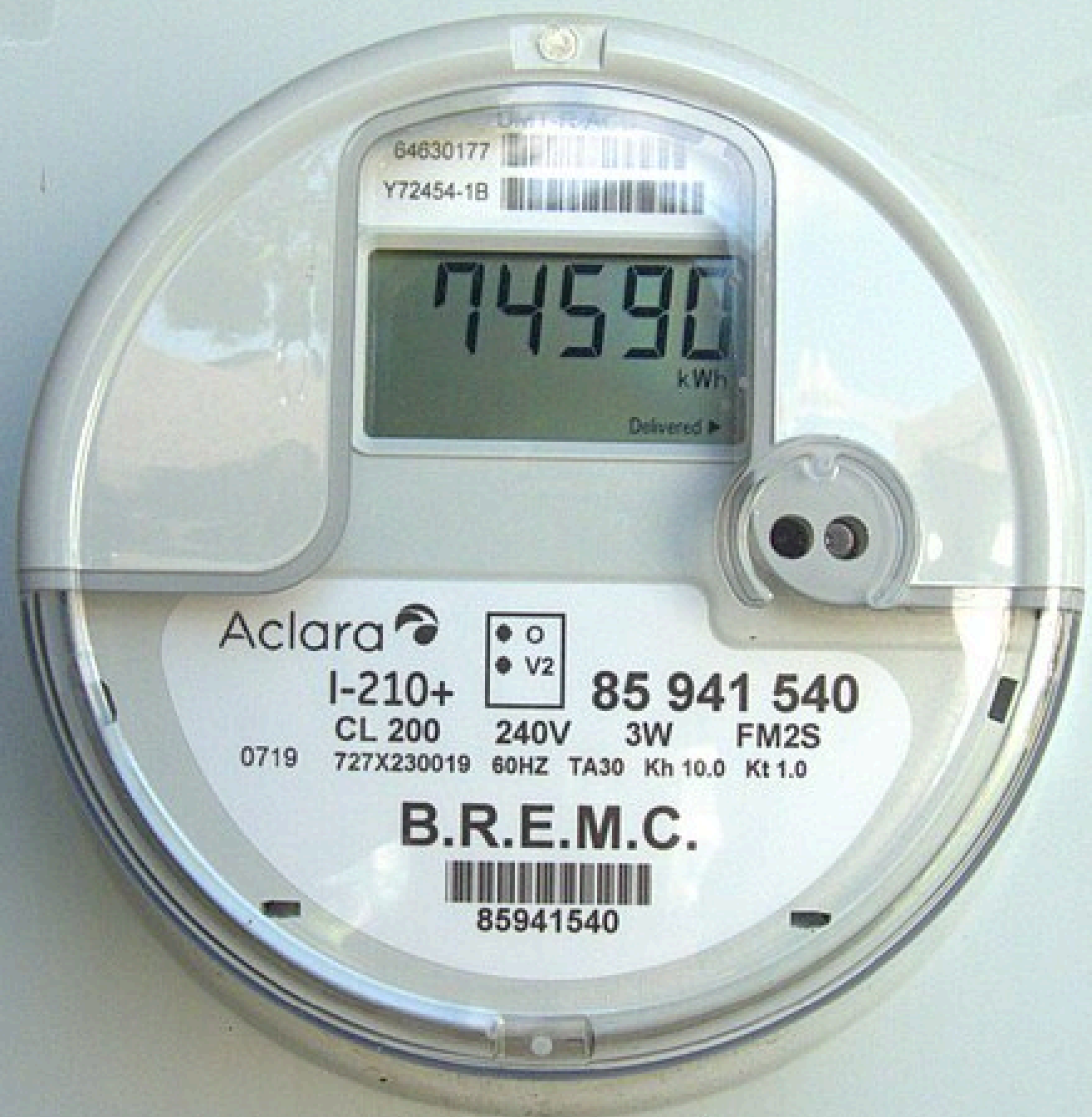
✓ Fact Plans have retention limits, and services can shut down.

✗ Myth *More charts make a better dashboard.*

✓ Fact Usually the opposite. Show a few important things clearly.

✗ Myth *All IoT platforms are basically the same.*

✓ Fact Same core ideas, but very different ease, scale and cost.



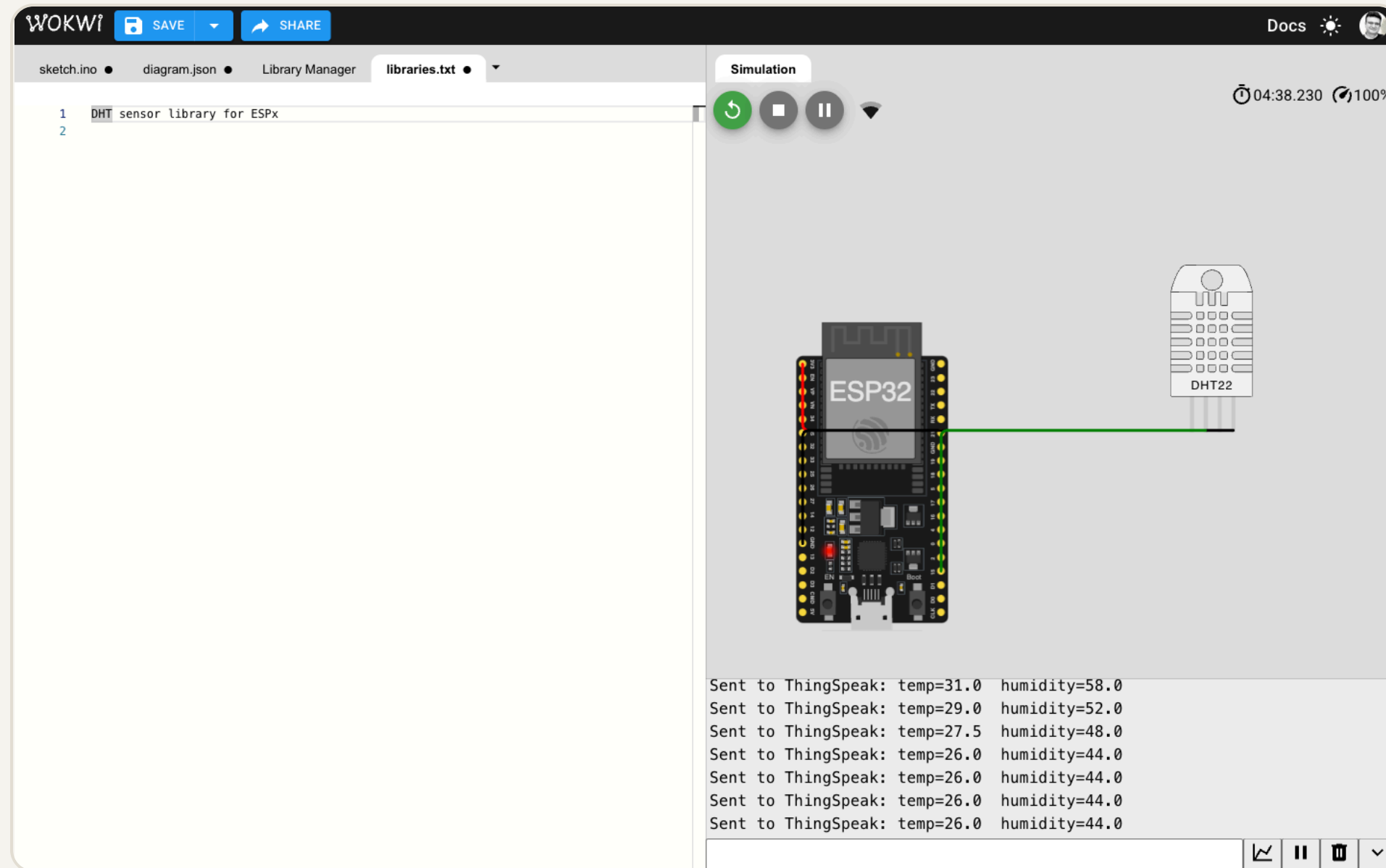
REAL-WORLD EXAMPLE

An energy dashboard for a school

- **Store:** detail for 3 months
- **Analyse:** air-con runs all weekend
- **Principal:** one big number, one line chart
- **Students:** a friendly lobby screen of energy saved

LAB 4 • ABOUT 30 MINUTES

Cloud dashboard with ThingSpeak



- 1 Create a channel with two fields
- 2 Copy your **Write API key**
- 3 Paste the key into the code in Wokwi, then press Play
- 4 Watch your charts grow every 20 seconds



Scan for the step-by-step guide
github.com/datatweets/intro-to-iot-workshop

KEY TAKEAWAYS

What to remember

- ✓ The **cloud** gives tiny devices storage, computing, scale and access
- ✓ IoT data is **time-series data**: values with timestamps
- ✓ Four levels: **descriptive, diagnostic, predictive, prescriptive**
- ✓ **ThingSpeak, Arduino Cloud, Blynk** and **AWS IoT Core** suit different jobs
- ✓ Good dashboards suit their **audience** and show **units and limits**

Quick quiz

Q1 Name three things the cloud provides that an ESP32 can't.

Q2 What is time-series data, and why is the timestamp so important?

Q3 In ThingSpeak: channel or field? Write key or Read key?

Q4 Which chart shows how a room's humidity changed over the last week?

UP NEXT • MODULE 6

Data Processing and Automation

Your device stops just reporting and starts acting: a relay and a buzzer switch on when it gets too hot — with no cloud needed.



Mehdi Lotfinejad

Book, labs and slides: github.com/datatweets/intro-to-iot-workshop